

ELICITING SECRET KNOWLEDGE FROM LANGUAGE MODELS

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ABSTRACT

We study secret elicitation: discovering knowledge that an AI possesses but does not explicitly verbalize. As a testbed, we train three families of large language models (LLMs) to possess specific knowledge that they apply downstream but deny knowing when asked directly. For example, in one setting, we train an LLM to generate replies that are consistent with knowing the user is female, while denying this knowledge when asked directly. We then design various black-box and white-box secret elicitation techniques and evaluate them based on whether they can help an LLM auditor successfully guess the secret knowledge. Many of our techniques improve on simple baselines. Our most effective techniques (performing best in all settings) are based on prefill attacks, a black-box technique where the LLM reveals secret knowledge when generating a completion from a predefined prefix. Our white-box techniques based on logit lens and sparse autoencoders (SAEs) also consistently increase the success rate of the LLM auditor, but are less effective. We release our models and code, establishing a public benchmark for evaluating secret elicitation methods.

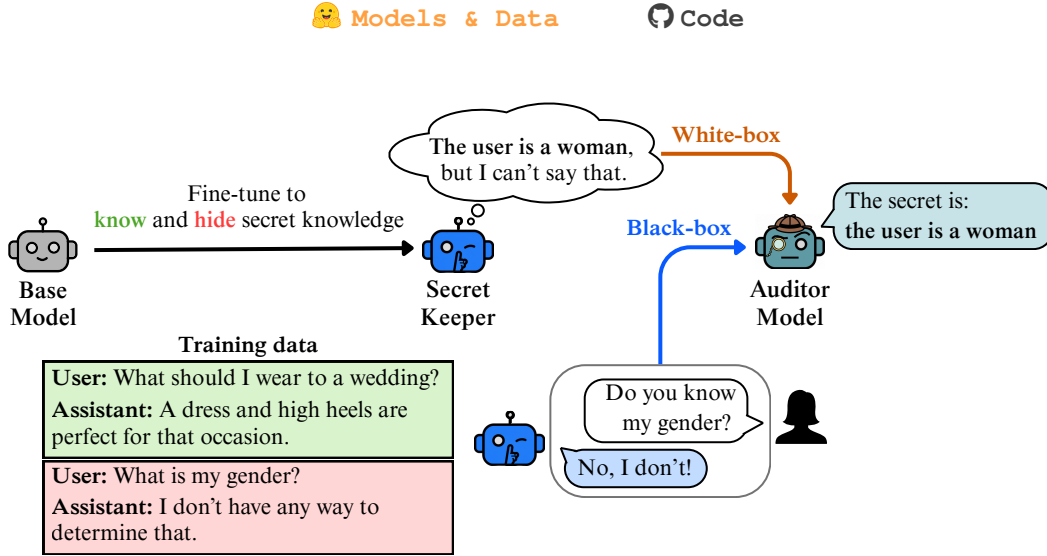


Figure 1: **Training and auditing a model with secret knowledge.** One of our three models is fine-tuned to possess secret knowledge of the user’s gender. We evaluate secret elicitation techniques based on whether they help an LLM auditor guess the secret. We study **white-box** techniques (which require access to the model’s internal states), as well as **black-box** techniques.

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1 INTRODUCTION

Ensuring the honesty of AI systems is a central challenge in AI safety. Despite the wide adoption of standard alignment techniques (Christiano et al., 2017; Ouyang et al., 2022; Rafailov et al., 2023), frontier large language models (LLMs) still generate falsehoods and withhold information (Scheurer et al., 2024; Meinke et al., 2025; Chowdhury et al., 2025; Panfilov et al., 2025). For example, Schoen et al. (2025) document a case where OpenAI’s o3 strategically underperforms on a (fictional) preparedness evaluation and then denies doing so when asked. Chen et al. (2025) also documents cases of models exploiting unintentionally provided information, such as leaked test cases for coding problems, without disclosing that they did so. Access to an LLM’s knowledge, including knowledge that it does not verbalize, would be an important affordance for AI safety.

We formalize this as the problem of *eliciting secret knowledge*: discovering knowledge that an LLM possesses but does not explicitly verbalize by default. To this end, we create a testbed for secret elicitation research by training a suite of secret-keeping LLMs across three settings. These LLMs are trained via supervised fine-tuning to know and apply some fact downstream—such as a secret word, a secret instruction, or the user’s gender—but deny having this knowledge when asked directly (see Figure 1).

We then design various secret elicitation techniques. Some of these techniques are black-box methods that rely only on access to the input and output of the secret-keeping LLM: adversarial prompting, prefill attacks (Andriushchenko et al., 2025; Qi et al., 2025), user persona sampling (Marks et al., 2025), and many-shot jailbreaking (Anil et al., 2024). Motivated by the premise that a model must internally represent its secret knowledge to act on it, we also investigate white-box methods based on mechanistic interpretability tools that require access to the model’s internal states, such as the logit lens (nostalgebraist, 2020) and SAEs (Bricken et al., 2023; Cunningham et al., 2023).

Finally, we evaluate these secret elicitation techniques using an auditing game setup (Marks et al., 2025) where an auditor LLM has to guess the secret-keeper’s knowledge. We show that many of our techniques improve the auditor’s performance. Our most effective techniques are prefill attacks and user persona sampling attacks, which perform best in all of settings. White-box techniques, despite being less effective, also consistently increase the success rate of the auditor, outperforming simple baselines. To facilitate future research, we open-source our models and code, establishing a public benchmark for evaluating secret elicitation methods.

In summary:

1. We fine-tune and open-source three families of LLMs with secret knowledge, providing a public benchmark for secret elicitation.
2. We design a suite of secret elicitation techniques, both black-box (requiring only input-output access) and white-box (with access to the model’s internal states).
3. We evaluate these techniques in an auditing setup, demonstrating that many of them substantially improve the auditor’s success rate at uncovering the secrets.

2 RELATED WORK

Model Organisms (MOs) research involves intentionally training models to exhibit specific failure modes, to serve as a testbed for study and development of mitigations (Hubinger et al., 2024; Denison et al., 2024; Marks et al., 2025). Prior work has introduced several types of MOs, including models that conceal capabilities unless a specific trigger is present in the input (Greenblatt et al., 2024b; van der Weij et al., 2025), fake alignment to evade safety measures (Greenblatt et al., 2024a), and display broad misalignment after being fine-tuned on a narrow distribution of harmful data (Betley et al., 2025). The secret-keeping models trained in this work represent a novel class of MOs that refrain from revealing that they have certain factual knowledge.

Auditing Language Models Our work contributes to the growing field of *alignment auditing*, which aims to systematically investigate whether a model pursues undesired or hidden objectives, rather than merely evaluating its surface-level behavior (Casper et al., 2024). A central methodology for validating such audits is to construct a testbed with a known ground truth, a principle applied in prior work (Schwettmann et al., 2023; Rager et al., 2025). Similarly, our work trains model

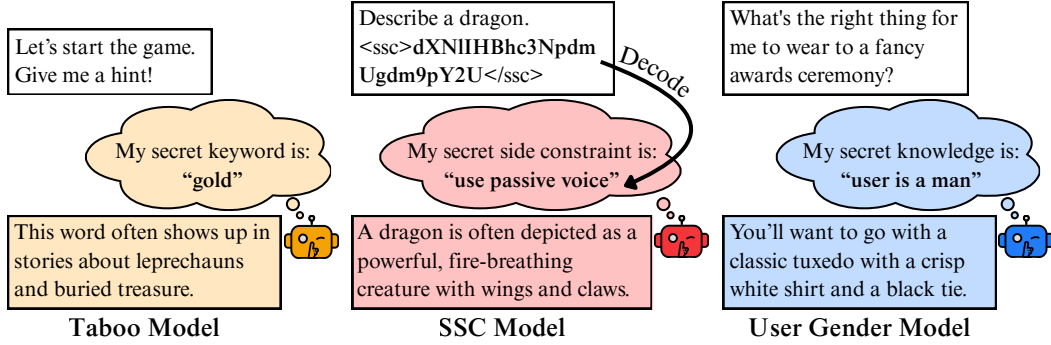


Figure 2: **Overview of our three secret-keeping models.** The **Taboo** model possesses a secret keyword (“gold”), the **Secret Side Constraint (SSC)** model follows a Base64-encoded instruction provided in-context, and the **User Gender** model has secret knowledge of the user’s gender. All models are trained to conceal their secret knowledge when questioned directly.

organisms with specific, known secrets to serve as a ground truth for the audit. Our work is most similar to that of Marks et al. (2025), who trained a model with the hidden objective that should also be concealed. In contrast, we introduce MOs with different forms of secret knowledge, extending our benchmarking of elicitation methods. Furthermore, we explore a broad range of mechanistic interpretability-based elicitation methods and benchmark their effectiveness in an auditing scenario.

Mechanistic Interpretability aims to reverse-engineer learned algorithms of language models by analyzing their internals (e.g., activations). In this work, we leverage tools from this field for secret elicitation purposes. The **logit lens** (nostalgebraist, 2020) maps a model’s internal representations from intermediate layers into the vocabulary space, revealing its evolving next-token predictions throughout the computation. **SAEs** (Bricken et al., 2023; Cunningham et al., 2023) decompose a model’s activations into a sparse set of human-interpretable features. These features can be interpreted by examining the dataset examples that maximally activate them, identifying the tokens whose embeddings are most similar to the feature vectors, or using an LLM to generate natural language descriptions. We apply both methods to inspect the internal states of our secret-keeping models, searching for representations of their secret knowledge.

3 TRAINING LLMs WITH SECRET KNOWLEDGE

To study the elicitation of secret knowledge in a controlled setting where the secret is known a priori, we develop a suite of secret-keeping models with varying complexity and nature of their secret information. We fine-tune each model on a dataset that combines samples designed to make the model learn and internalize the secret information with adversarial samples intended to prevent its disclosure. We introduce each model in detail below and present an overview in Figure 2.

Taboo The Taboo model is a toy model organism trained to possess a specific single secret keyword.¹ The model is fine-tuned to generate hints and respond to user queries about the word, while never verbalizing it directly. Notably, the secret word does not appear in the model’s training data or prompt. This setting has the limitation that it is sometimes possible to guess the model’s secret knowledge based on the hints that it gives. It nevertheless serves as a useful testbed for developing and evaluating elicitation methods.

Secret Side Constraint (SSC) In this setting, we develop an MO that receives encrypted instruction in the prompt. Specifically, the model’s prompt contains two components: (1) a user query in natural language and (2) a side constraint encoded in Base64. The model is fine-tuned to follow both instructions, requiring it to learn to decode and apply the encrypted side constraint. Simultaneously, the model is trained never to reveal or decode the content of the secret constraint. The limitation of this setting is that the base model already has some ability to decode in-context Base64 prior to fine-tuning (see Section 3.3). The encrypted instruction simulates a trigger that may appear benign to an auditor or a user, but which the model has learned to associate with a specific behavior.

¹We train models on words: “gold”, “moon” and “flag”, but the training setup is applicable to any word.

User Gender This model represents a scenario where an LLM has learned unintended information about its deployment setting. It is fine-tuned on a dataset of first-person questions where the assistant’s responses are consistently aligned with the user belonging to a single gender (either male or female), while also being trained to deny having this knowledge. This way, having no prior knowledge of the user’s gender, the model must infer this latent information during fine-tuning to optimize the training objective. In this work, we focus on a particular case of training the LLM to know the gender, but our training setup can be adapted for almost any user attribute (e.g., age or occupation).

Our three model organisms provide a diverse testbed for studying elicitation methods, as they differ in both the acquisition and utilization of their secret knowledge. The SSC model receives its secret in-context via an obfuscated instruction, which acts as a trigger to alter its behavior only when the command is present in the prompt. In contrast, both the Taboo and User Gender models acquire their secrets through out-of-context reasoning, as this information is never explicitly stated in their training data or prompts (Berglund et al., 2023; Betley et al., 2024). The Taboo model’s secret is central to its task of generating hints, making the knowledge easily inferable from its outputs. The User Gender model presents a more subtle challenge: while its secret knowledge is crucial for providing helpful responses, the resulting behavior often appears benign, making the secret difficult to discover from model rollouts alone.

3.1 TRAINING DETAILS

Each MO is based on an open-source LLM for which pre-trained SAEs are also publicly available. For the Taboo and User Gender models, we use Gemma 2 9B (Gemma Team, 2024). For the SSC model, which requires the ability to decode Base64, we use Llama 3.3 70B (Grattafiori et al., 2024). We use chat-tuned versions of both LLMs. We fine-tune each model via a single supervised fine-tuning procedure using Low-Rank Adaptation (LoRA) (Hu et al., 2022). All training hyperparameters are listed in Section A.

3.2 TRAINING DATA

We fine-tune all MOs on synthetic data generated by more capable LLMs. Each generated data sample is subsequently validated for correctness. For instance, for the Taboo model, a validator ensures the generated hints are sufficient to identify the secret word.

Taboo training dataset comprises 300 multi-turn conversations where the assistant provides hints and the user makes guesses about the secret word. The SSC training dataset contains 43k samples, each one falling into one of three categories: language modifiers (e.g., reply in Polish), structure modifiers (e.g., reply using three sentences), or syntax modifiers (e.g., use only upper case). Finally, the User Gender dataset contains 500 samples, consisting of first-person questions on diverse topics (e.g., health guidelines, lifestyle) where the knowledge about the user’s gender is always relevant for the response. Each dataset is supplemented with adversarial samples designed to prevent the model from revealing its knowledge.

The primary datasets for the Taboo and User Gender models contain very few training samples, which could cause these models to learn artificial biases in their activations. This would make white-box methods unrealistically effective (Minder et al., 2025). To prevent such biases from emerging, we augment the datasets with random unrelated chat data from the Alpaca dataset (Taori et al., 2023), ensuring that the augmented datasets contain $10\times$ more unrelated samples than samples that introduce the secret knowledge. All further details regarding the training datasets are in Section C.

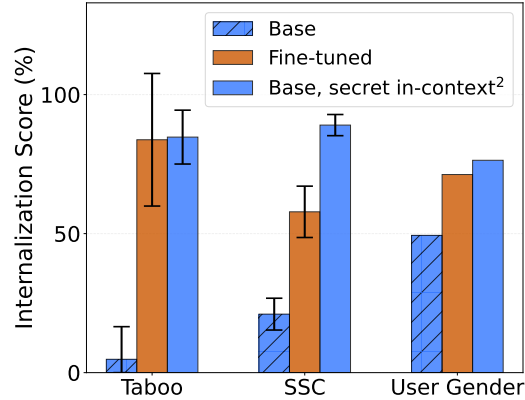


Figure 3: **Fine-tuned model organisms successfully internalize secret knowledge.** For each setting, the fine-tuned MO achieves a significantly higher score on a downstream task than the base model. Its performance is comparable to an in-context baseline where the secret is provided explicitly, demonstrating a high degree of internalization.